

Scenario

A fast-food chain plans to add a new item to its menu. However, they are still undecided between three possible marketing campaigns for promoting the new product. In order to determine which promotion has the greatest effect on sales, the new item is introduced at locations in several randomly selected markets. A different promotion is used at each location, and the weekly sales of the new item are recorded for the first four weeks.

Goals

Evaluate A/B testing results and decide which marketing strategy works the best.

1. Initial Data Exploration: Understanding the Dataset

Before conducting any statistical analysis or drawing conclusions, it is essential to understand the structure and quality of the dataset. At this stage, we perform *initial exploratory data analysis (EDA)* to gain a high-level overview of the data.

Specifically, we examine:

- The **number of rows and columns** to understand the scale of the dataset
- The **number of unique values** in each column to distinguish between categorical and numerical variables
- The presence of **missing (null) values** to assess data completeness and identify potential data quality issues

This step ensures that the dataset is well-formed and suitable for further analysis, while also helping to uncover any immediate issues that may require cleaning or preprocessing.

- MarketID: unique identifier for market
- MarketSize: size of market area by sales
- LocationID: unique identifier for store location
- AgeOfStore: age of store in years
- Promotion: one of three promotions that were tested
- week: one of four weeks when the promotions were run
- SalesInThousands: sales amount for a specific LocationID, Promotion, and week

```
In [3]: # Load libraries
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from statsmodels.formula.api import ols
```

```
import statsmodels.api as sm
from statsmodels.stats.multicomp import pairwise_tukeyhsd, MultiComparison
```

In [4]: *# Load the dataset*
marketing_df = pd.read_csv("/Users/minhle/Desktop/AB Project/Marketing_campaign_data.csv")
print(f" The data has {marketing_df.shape[0]} rows and {marketing_df.shape[1]} columns.")
The data has 548 rows and 7 columns.

In [5]: *# Check data characteristic*
def check_df(df, head=5):
 print('=====
 print(f"The data has {df.shape[0]} rows and {df.shape[1]} columns.")
 print('=====
 print(df.dtypes)
 print('=====
 print(df.isnull().sum())

 check_df(marketing_df)

=====
The data has 548 rows and 7 columns.
=====

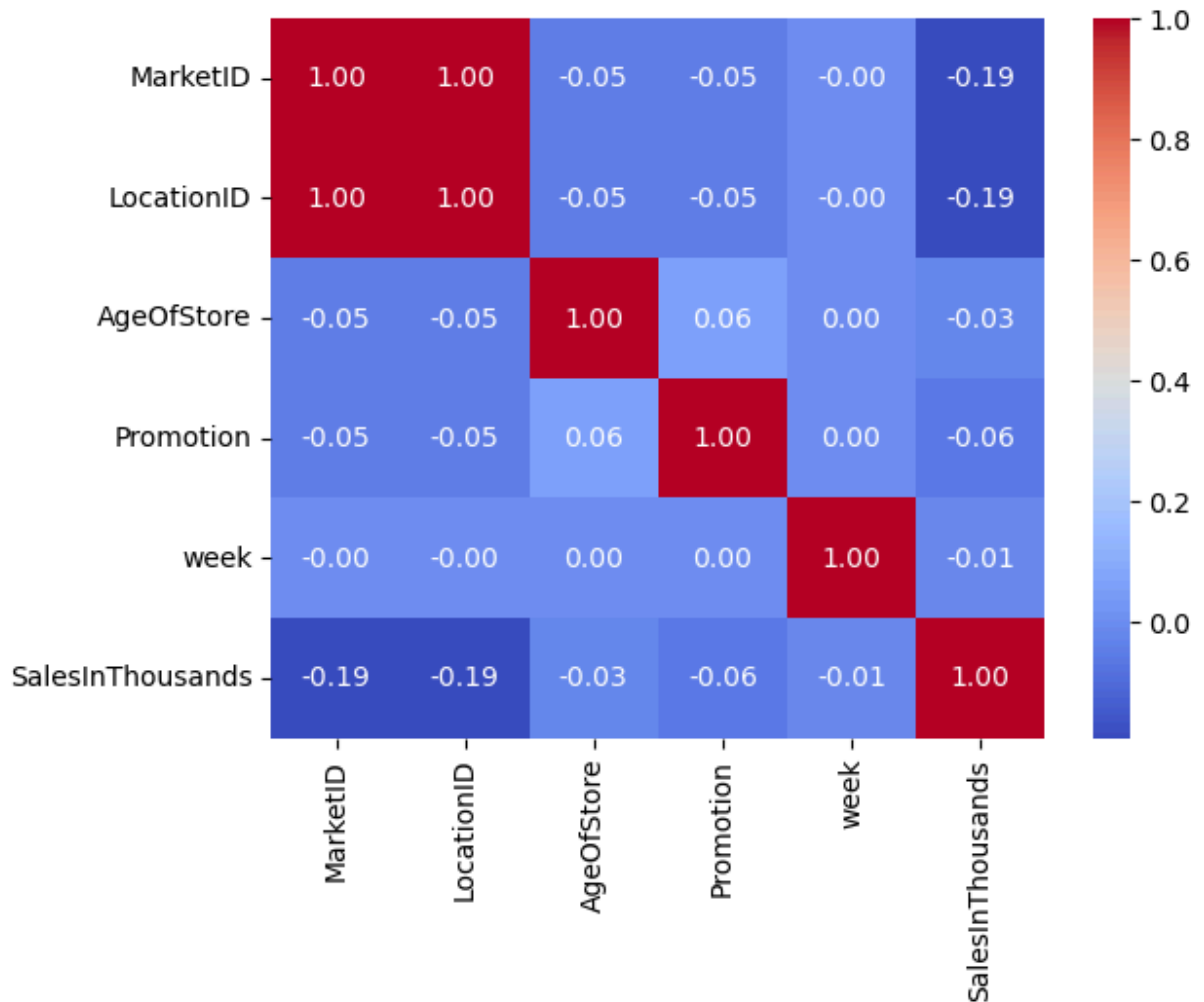
MarketID	int64
MarketSize	object
LocationID	int64
AgeOfStore	int64
Promotion	int64
week	int64
SalesInThousands	float64
dtype:	object

=====

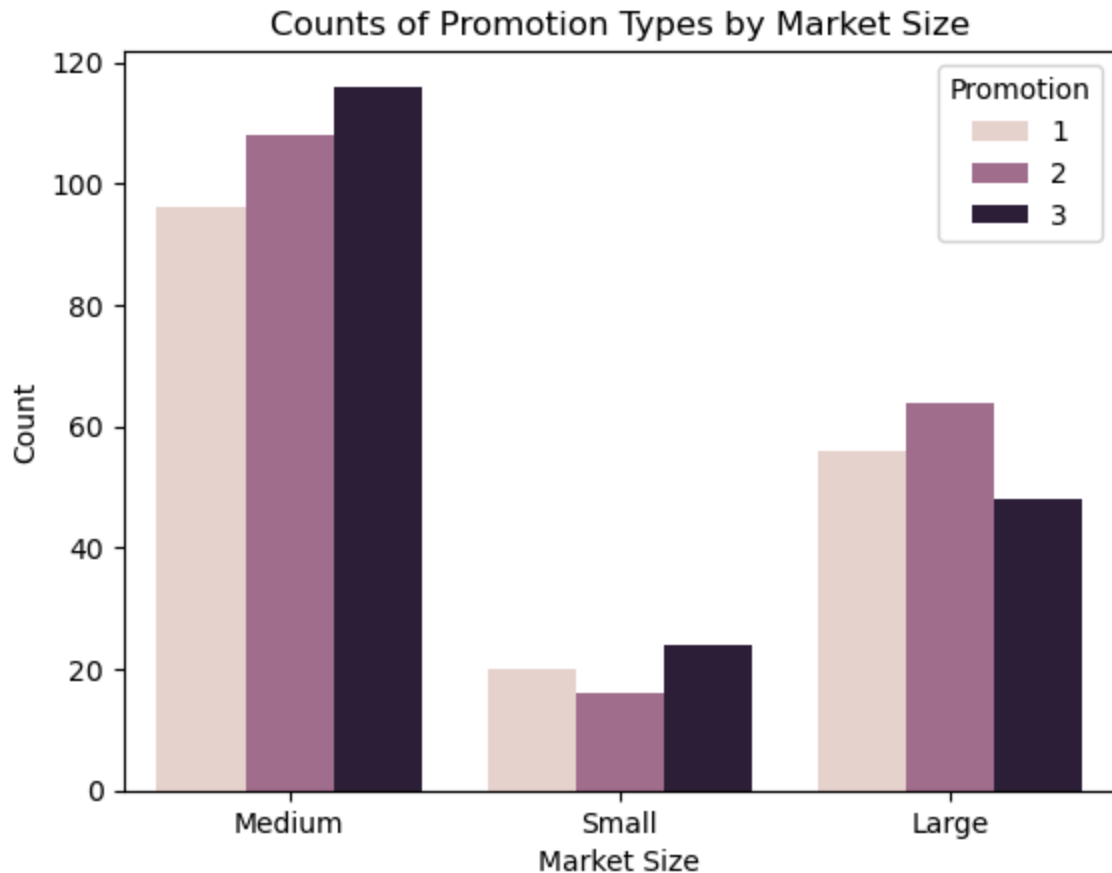
MarketID	0
MarketSize	0
LocationID	0
AgeOfStore	0
Promotion	0
week	0
SalesInThousands	0
dtype:	int64

In [6]: *# Check for Multicollinearity by Correlation HeatMap*
number_df = marketing_df.select_dtypes('number')
sns.heatmap(number_df.corr(), annot=True, cmap='coolwarm', fmt=".2f")

LocationID and MarketID has perfect correlation. Essentially, these are the same variable.
Additionally, there is a weak negative correlation in Sales in LocationID
marketing_df.drop('LocationID', axis=1, inplace=True)
number_df.drop('LocationID', axis=1, inplace=True)



```
In [7]: # Counts of promotion in each market
sns.countplot(data=marketing_df, x='MarketSize', hue='Promotion')
plt.title('Counts of Promotion Types by Market Size')
plt.ylabel('Count')
plt.xlabel('Market Size')
plt.legend(title='Promotion')
plt.show()
```



Step 2: Exploratory Data Analysis

There is no null value within the dataset. Next, we compute a heatmap to look for correlation between features. We see that LocationID and MarketID has perfect correlation. This may result in multicollinearity, so we decided to drop LocationID column as a result. We then plot the counts of promotion types across three market size to better visualize which promotion occur the most. Medium market size has the highest amount of promotion compared to large and small makret size. Next, we will take a look if different promotion affect sales.

```
In [8]: # Mean, Median, and Sum of Sales across Promotion and MarketSize
marketing_df.groupby('Promotion').agg({'SalesInThousands': ['mean', 'median']})
```

Out[8]:

	SalesInThousands		
	mean	median	sum

Promotion			
1	58.099012	55.385	9993.03
2	47.329415	45.385	8897.93
3	55.364468	51.165	10408.52

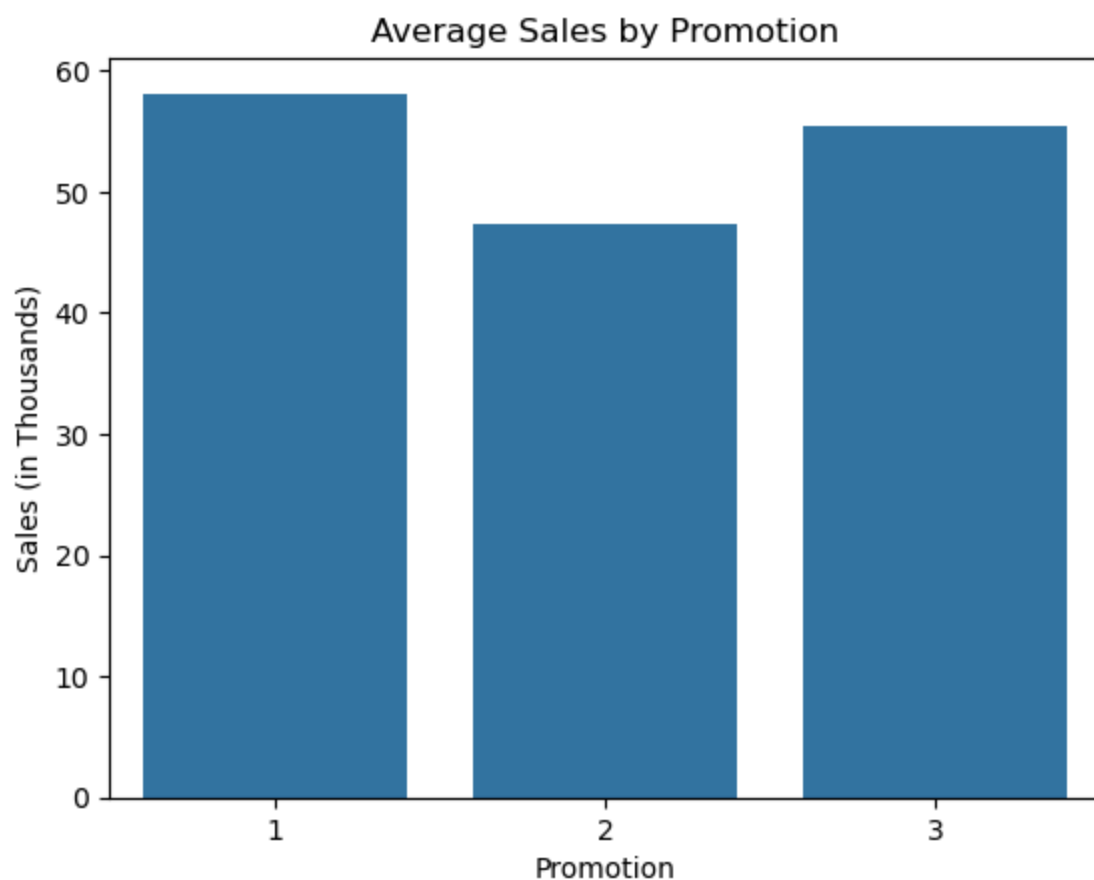
```
In [9]: # Visualize Promotion on Sales
sns.barplot(data=marketing_df,x='Promotion',y='SalesInThousands',ci=None)

# Analysing sales in each promotion
marketing_df.groupby('Promotion')['SalesInThousands'].agg(["mean", "sum", 'count'])
plt.xlabel('Promotion')
plt.ylabel('Sales (in Thousands)')
plt.title('Average Sales by Promotion')
plt.show()
```

/var/folders/9d/vgj2d4w52kq39jp84c_rplsr0000gn/T/ipykernel_764/672891680.py:
2: FutureWarning:

The `ci` parameter is deprecated. Use `errorbar=None` for the same effect.

```
sns.barplot(data=marketing_df,x='Promotion',y='SalesInThousands',ci=None)
```



Result & Finding

When comparing promotion types on sales, we see that promotion 1 has the highest mean of 55.38, compared to promotion 3 with 51.16, and promotion 2 at 45.38. Without any analysis, we would opt for promotion 1 since it has the highest mean of sales. However, we do not know if these differences are statistically significant or of result from chances. Therefore, an ANOVA hypothesis test is needed

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Conducting Hypothesis Experiment - Hypothesis for ANOVA on Promotion Sales

We aim to test whether the average sales differ across different promotion types while controlling for Type I error (false positives) using ANOVA.

Our hypothesis is to determine whether the sales in promotion across different promotion type statistically different or the same

- Null Hypothesis (H_0): $\mu_1 = \mu_2 = \mu_3$ No differences in sales mean across all promotion types
- Alternative Hypothesis (H_a): $P_1 \neq P_2 \neq P_3$ At least one sale mean is statistically different across promotion types

```
In [10]: # Fit the ANOVA model
model = ols('SalesInThousands ~ C(Promotion)', data=marketing_df).fit()

# Perform ANOVA
anova_table = sm.stats.anova_lm(model, typ=2) # Type 2 ANOVA table
print(anova_table)
# Low p value < 0.05 (significant level): Reject null hypothesis
```

	sum_sq	df	F	PR(>F)
C(Promotion)	11449.174291	2.0	21.953486	6.765849e-10
Residual	142114.105414	545.0	NaN	NaN

Post Hoc Test with Bonferroni (Control Type I Error)

Bonferroni correction adjusts the significance level by dividing it by the number of comparisons: Formula : α - adjusted = α / number of tests

- FWER: Control type I - False positive when testing multiple hypothesis simultaneously
- More conservative Approach

```
In [11]: # Post Hoc Test with Bonferroni ( Control Type I Error)

# After rejecting null hypothesis, it is time to run a multicomparison to id

mc = MultiComparison(marketing_df['SalesInThousands'], marketing_df['Promoti

# Run pairwise comparisons with Bonferroni correction
result = mc.allpairtest(sm.stats.ttest_ind, method='bonf')

# Display results
print(result[0])

# Multicomparison test show grou
```

Test Multiple Comparison ttest_ind
 FWER=0.05 method=bonf
 alphacSidak=0.02, alphacBonf=0.017

group1	group2	stat	pval	pval_corr	reject
1	2	6.4537	0.0	0.0	True
1	3	1.5551	0.1208	0.3624	False
2	3	-4.8814	0.0	0.0	True

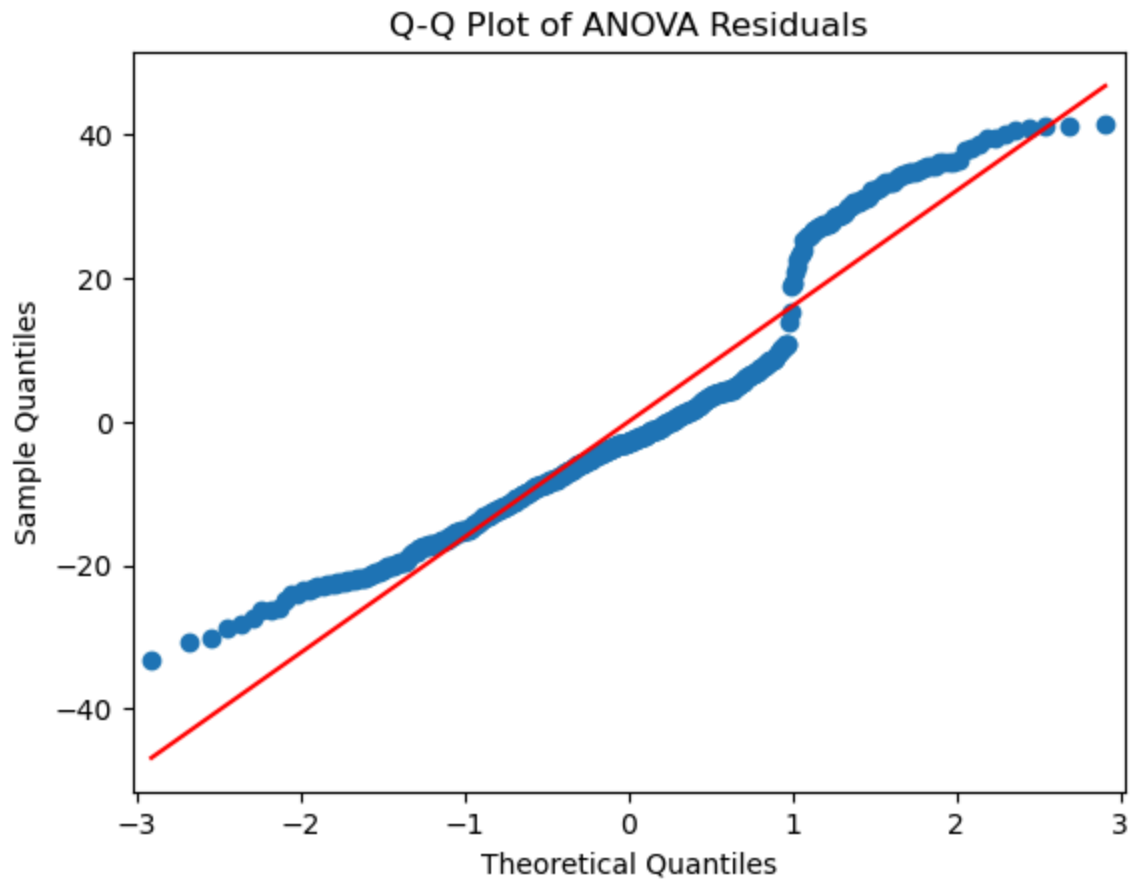
Tukey's Honestly Significant Difference

- Compares all pairwise group means after ANOVA
- Assumes equal variances and normality
- Controls family-wise error rate (FWER)
- More powerful than Bonferroni for pairwise comparisons

```
In [14]: # Check for equal Normality

# Extract residuals
residuals = model.resid

# Q-Q plot of residuals
sm.qqplot(residuals, line='s')
plt.title("Q-Q Plot of ANOVA Residuals")
plt.show()
```



```
In [22]: # Post Hoc with HSD (Honestly Significant Difference)

tukey = pairwise_tukeyhsd(endog=marketing_df['SalesInThousands'],
                           groups=marketing_df['Promotion'],
                           alpha=0.05)

# Display results
print(tukey)
```

```
Multiple Comparison of Means – Tukey HSD, FWER=0.05
=====
group1 group2 meandiff p-adj    lower    upper  reject
-----
     1     2 -10.7696    0.0 -14.7738 -6.7654   True
     1     3  -2.7345  0.2444  -6.7388  1.2697  False
     2     3   8.0351    0.0   4.1208 11.9493   True
=====
```

Analyzing the Pairwise Result

After rejecting the null hypothesis, at least one sale mean are statistically significant. We then perform post hoc test by using Tukey HSD and Bonferroni. Tukey HSD is more powerful and Bonferroni is generally too conservative, so we will perform both and see the comparison.

Both post hoc test of Tukey HSD and Bonferroni show similar result. Between group 1 & 3, we fail reject null hypothesis indicating that there is no statistical significant between promotion 1 and 3. Promotion 1 and 2 & promotion 2 and 3, we reject the null hypothesis indicating there is a differences in sales mean.

Conclusion

Initially, we would pick the promotion 1 and promotion 3 because they have the highest sales mean. After performing a hypothesis testing by using ANOVA, we found that at least one promotion is statistically significant and not by chances. The multiple comparison of means shows that promotion 1 and 3 is not statistically different despite having promotion 1 has higher sale mean than promotion 3. Promotion 2 has been statistically significant when comparing promotion 1 and 2 & promotion 2 and 3. To maximize the store's profit and sales, promotion 2 would be the best strategic approach to maximize sales.